

Capacity Eviction in Reconstruction-Trained Tokenizers

Draft of August 9, 2026 — section text frozen pending review

Note to the reviewer. This is §3 of a paper in preparation, exported verbatim from the manuscript together with the three appendices it depends on. Numbering matches the manuscript: §3.1–§3.8 and Appendices A–C. Figure 1 (the architecture schematic) belongs to §1 and is not reproduced here; Figure 2 is final in content but has one further production pass pending, and Figure 3 is a *placeholder* standing in for a figure that has been specified but not yet built — its caption states what that figure will show. Everything else, prose and numbers, is as intended for submission.

What we would most like challenged. (i) Whether the boxed Claim in §3.1 is stated at a defensible scope — as a property of reconstruction objectives in general, or of this tokenizer class under this loss; the evidence is synthetic-fixture evidence and we may have over-reached. (ii) Whether §3.2’s fixture genuinely isolates the single property it claims to isolate. (iii) Whether the token-space control in §3.4 licenses the localization conclusion, or whether a confound survives it. (iv) Any number that does not follow from what is stated.

3 Capacity eviction in reconstruction-trained tokenizers

Two-stage models for financial time series learn a tokenizer under a reconstruction objective and then fit an autoregressive model to the resulting discrete codes [Shi et al., 2025, van den Oord et al., 2017]. The design carries an assumption that is seldom stated: that *a representation good enough to reconstruct from is good enough to forecast from*. The tokenizer is fitted to reconstruction, the forecaster never sees the underlying series, and nothing in the pipeline is trained to preserve what the forecaster will need.

We show that the assumption fails, that it fails deterministically rather than by chance, and that it fails hardest on precisely the channels a microstructure-aware model depends on. We state the mechanism (Section 3.1), measure it (Section 3.2), test the prediction it makes about an end-to-end pipeline (Section 3.3), localize the loss by construction rather than by inference (Sections 3.4 and 3.5), give an intervention that removes it (Section 3.6), and convert the result into a standing gate that must pass on real data before any second-stage expenditure (Section 3.7). Every measurement in this section is made on a synthetic fixture with a known ground-truth signal, chosen so that the quantity being recovered is exactly computable; Section 3.8 states what that does and does not license.

3.1 The mechanism

A tokenizer maps each bar to a code of fixed size, so capacity is scarce and must be allocated across the bar’s dimensions. The allocation is chosen by whatever the objective rewards, and a reconstruction objective rewards two things.

Claim. Under a reconstruction objective, code capacity is allocated in order of the loss reduction each dimension buys. That ordering is set by a dimension’s *variance*, which bounds how much loss it can contribute, and by its *covariance* with dimensions already coded, which determines how cheaply it can be served by capacity already spent. Downstream predictive value enters nowhere. A dimension that is low-variance *and* weakly covariant with the rest is therefore the last claim on the budget. Where two conditions hold together — a budget tight enough that not every dimension can be coded, and competing dimensions correlated enough that coding them is partly amortized — that dimension receives nothing, and does so deterministically: not occasionally, but in every run. Where either condition fails, the same objective still allocates by the same rule, but the exclusion is no longer determined; Section 3.2 reports a fixture in which the second condition is absent and the allocation is an initialization lottery instead.

Table 1: Reconstruction quality by variance class, three seeds. Point-decoder correlation between a dimension’s true value at bar t and its reconstruction from bar t ’s own fine code. Every dimension except the return has unit marginal variance; the signal dimension differs from the fillers only in being statistically independent. Filler entries give the range over the eleven filler dimensions; per-dimension values are in Appendix A.

dimension class	cross-dimensional structure	seed 0	seed 1	seed 2
return, sd 3.15	driven by the signal at lag 2	0.9818	0.9815	0.9823
fillers (11), sd 1.00	AR(1) chain across dims, $\rho = 0.7993$	0.826–0.918	0.824–0.919	0.824–0.917
signal, sd 1.00	independent	0.0140	0.0011	0.0053

The budget is tight, and the arithmetic says by how much. The canonical configuration uses per-dimension levels (11, 9, 9, 7, 7, 5, 5), split into a coarse subtoken of $11 \times 9 \times 9 = 891$ codes (9.80 bits) and a fine subtoken of $7 \times 7 \times 5 \times 5 = 1225$ codes (10.26 bits), for 20.058 bits per bar; the binary-spherical arm [Zhao et al., 2024] spends 10 + 10 bits for 20.000. Recovering the sign of a jointly-normal dimension at accuracy a requires a correlation $\rho = \cos(\pi(1 - a))$, so the $a = 0.90$ we will require costs at least $-\frac{1}{2} \log_2(1 - \rho^2) = 1.69$ bits for that dimension alone by the Gaussian rate–distortion bound, and more under any realizable scalar quantizer. Thirteen live dimensions at that fidelity would need at least 22.0 bits — more than the entire per-bar budget, at any coarse/fine split. Not every dimension can be coded well. Something must be dropped, and the objective decides what.

This matters here because microstructure has exactly the profile the claim identifies. Trade-flow imbalance and the aggregate trade-flow statistics are low-variance relative to returns and only weakly covariant with price shape — they carry information about the near future precisely because they are *not* a restatement of the recent past. A tokenizer trained to reconstruct will therefore allocate capacity away from them by construction. This is not a defect of the objective; it is the objective working correctly against a goal it was never given. It also means that “microstructure-aware” has to be an architectural property of a tokenizer, and cannot be an emergent one.

3.2 Measuring the eviction

To measure the claim we need a fixture in which one dimension differs from the others in exactly the property the claim names, and in nothing else. We construct a synthetic bar stream in which every dimension is marginally standard normal — so all of them contribute equally to a variance-weighted loss — and then make eleven of them mutually dependent while one, the *signal* dimension, is independent of everything.

The dependence is not chosen for convenience; it is calibrated to the real data. On 30 symbols and 3,895,808 unmasked bars, the Gaussian-copula correlation log-determinant over the seven non-microstructure dimensions is -6.4819 , a per-dimension entropy deficit of 0.4631 nats. The fixture reproduces that deficit with a cross-dimension AR(1) chain at $\rho = 0.7993$, achieving 0.462 nats against a 0.05-nat tolerance. One further dimension, the return, is driven by the signal dimension at a two-bar lag and so carries 3.15 $\times$ the marginal standard deviation of the rest: it is the high-variance channel, and it is also the only dimension with downstream value in this fixture, since it is what a forecaster would predict.

Figure 2 and Table 1 report, for three independently initialized tokenizers, how well each dimension can be recovered from the bar’s own code. The result sorts by variance class and does not vary across seeds. The return is recovered at 0.98. The eleven mutually dependent dimensions are recovered at 0.82–0.92, because one shared component serves all eleven at once and the capacity spent on it is amortized eleven ways. The signal dimension — same marginal variance as any single filler, differing only in being independent — is recovered at 0.001–0.014. It is not coded at all.

Calibrating the fixture to the real data is what makes this a mechanism rather than an anecdote, and we report the correction it forced because it changes the strength of the claim. On an earlier fixture whose dimensions were exchangeable i.i.d. draws — no cross-dimensional dependence at all — the allocation was arbitrary: across three initializations the set of dimensions clearing 0.9 changed completely, and the dimension of interest cleared in one seed of three. That reading makes the phenomenon an initialization lottery, which is a substantially weaker and less actionable claim.

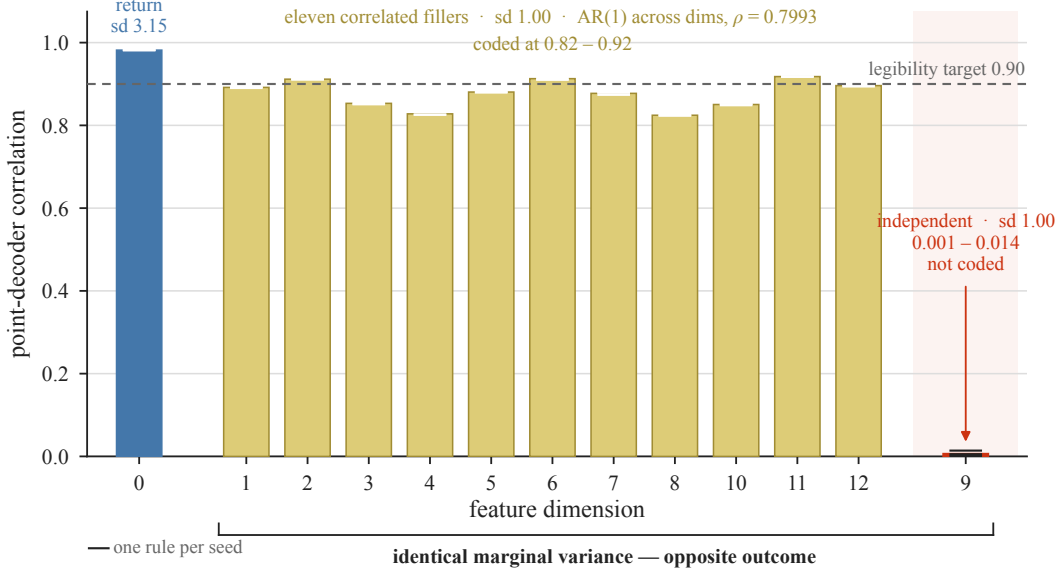


Figure 2: Reconstruction quality is sorted by variance class, not by task relevance, and the ordering is deterministic. Point-decoder correlation between each dimension’s value at bar t and its reconstruction from bar t ’s own code, for three independently initialized tokenizers on a fixture whose non-signal structure is entropy-calibrated to the real data. The return dimension, whose marginal standard deviation is $3.15\times$ that of the others, is recovered at 0.98; four of the thirteen dimensions clear the 0.90 rule in every seed. The eleven filler dimensions, which have unit marginal variance and are coupled by an AR(1) chain across dimensions at $\rho = 0.7993$, are recovered at 0.82–0.92. The signal dimension has *the same unit marginal variance as the fillers* and differs from them in exactly one property — it is statistically independent of everything — and is recovered at 0.001–0.014. The three seeds agree to within the marker width, so this is deterministic exclusion rather than an initialization lottery. Reconstruction objectives buy variance and covariance; they do not buy independence.

Introducing the dependence structure the real data actually has replaced the lottery with deterministic exclusion, because the correlated block then offers an amortization the independent dimension cannot match.

3.3 The prediction, tested

If a tokenizer evicts a channel then a signal carried only by that channel is unavailable to everything downstream, however much information it contains and however capable the downstream model is. We test that prediction by planting a signal of known size.

We build a three-symbol stream of 42,000,000 bars. The signal dimension s_t is drawn i.i.d. standard normal and observed at the close of bar t ; in the *planted* arm it drives the return two bars later,

$$r_t = \sigma(\varepsilon_t + c s_{t-2}), \quad c = 3, \quad \varepsilon_t \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (1)$$

and in the *noise* arm s_t is redrawn independently, so the arms differ in exactly one respect. Because the states are i.i.d., past returns reveal only states at lag ≥ 2 , which are orthogonal to the forward label: the plant is recoverable *only* through the microstructure channel, and not through any function of price history. Equation (1) is a Gaussian channel, so the information it injects is exact,

$$I(s_t; r_{t+2}) = \frac{1}{2} \ln(1 + c^2) = \frac{1}{2} \ln 10 = 1.151 \text{ nats per bar}, \quad (2)$$

and because the tokens are a deterministic function of the features, the data-processing inequality makes 1.151 nats an upper bound on the validation cross-entropy improvement the autoregressive stage can obtain from the plant.

Both arms were trained under an identical budget and schedule — 20,000 steps at batch 32 and context 32, giving 20,480,000 bar-visits against a 27,300,000-bar training region, so 0.75 epochs,

and memorization is not available as an explanation. The sub-epoch condition is asserted in the run receipt rather than assumed.

The pipeline extracted none of it. The schedule completed on both arms with no loss spikes; the validation–train gap was +0.0072 (noise) and −0.0052 (planted), confirming the sub-epoch regime; the correlation between the model’s forecast and the planted state stayed below 0.05 at *every one of the 40 evaluation points*, with a maximum of 0.0433; and the planted arm’s final validation NLL was 13.6113 against the noise arm’s 13.5979 — worse by 0.0134 nats, which is the wrong sign.

The tokenizer was not discarding the state. Window-level reconstruction of the planted dimension was non-degenerate in the same run: MAE 0.3044 against a predict-the-mean baseline of 0.8216. The information was present in the code stream and unavailable to the model consuming it.

3.4 Localizing the loss: a token-space control

A null admits two explanations: the tokenizer’s code geometry hides the rule, or the backbone cannot learn thin conditionals at this scale. We separate them by construction rather than by argument, planting a signal of comparable size *directly in token space* — bypassing the tokenizer entirely — and asking whether the same backbone at the same budget recovers it.

The plant redraws a target digit from a discretized Gaussian centred monotonically on a source digit, with its width solved so the mutual information lands in a pre-registered band. It delivers 0.9003 nats exact over the full stream, agreeing with a Monte-Carlo plug-in estimate to 2.0×10^{-5} , with a marginal KL of 0.0048 and an entropy shift of −0.005 nats — so no information is smuggled in through the marginal — and an unplanted control returns -8.8×10^{-5} . Construction, calibration, and the retirement of a first, information-poor plant design are given in Appendix B. The carrier is the regenerated noise stream of Section 3.3 tokenized by that run’s frozen tokenizer, so token distribution, budget, schedule and validation protocol are unchanged, and that run’s noise arm supplies the reference $H_0 = 13.5979$.

The backbone recovers it almost completely: the probe crosses the pre-registered detection threshold of 0.30 at step 2,000 and reaches 0.9999, and the final validation NLL is 12.7483, that is $H_0 - 0.8496$ nats, or 94.4% of the planted information. Figure 3 places the two experiments side by side against their planted-information references. The backbone is not the limiting factor. The loss is at the tokenizer–backbone interface.

3.5 The interface defect, measured directly

The window encoder is causal but *contextual*: bar t ’s code is a function of bars $\leq t$. State belonging to bar t is therefore free to be distributed across the codes of every subsequent bar in the window. The window as a whole reconstructs; no individual code need carry its own bar’s features. Since the autoregressive stage consumes exactly the per-bar code sequence, anything that lives only in that spread-out form is unreachable.

The measurement is unambiguous. On 300,000 bars of the planted stream, tokenized by the frozen tokenizer of Section 3.3 with a 240,000/60,000 split, a logistic probe recovering $\text{sign}(s_t)$ from bar t ’s own coarse and fine digits achieves 0.5135 against a chance rate of 0.5, and an MLP regressing s_t on the same digits achieves a Pearson correlation of 0.0142 — while the same tokenizer on the same stream reconstructs that dimension at the window level at MAE 0.3044 against a 0.8216 baseline. The state is encoded, and per-bar illegible.

A stronger form of the same defect appears when the encoder is not causal at all, and we record it because it is why causality is enforced. Perturbing only future bars inside a tokenization chunk and counting how many past bars’ tokens flip gives a leakage rate of 41.75% coarse and 28.33% fine on a tokenizer trained on real data; the causal-encoder control measures exactly 0.0% on both, which is what establishes that the probe can read zero. Causal encoding is necessary for validity, and this section establishes that it is not sufficient for usability: it removes the leak forward and leaves the smear backward.

3.6 Intervention

If capacity is allocated by an objective that cannot see per-bar legibility, the remedy is to give the objective a term that can. We make three changes. Each was necessary; none was sufficient alone,

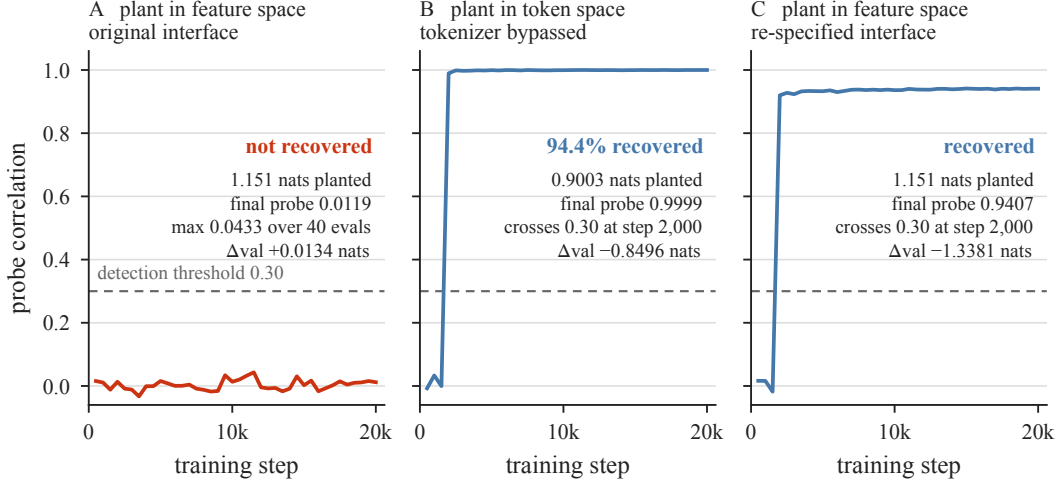


Figure 3: The same backbone extracts nearly everything from token space and nothing from feature space. Nats of validation cross-entropy recovered against training step, measured against a matched reference run, with the dashed rule in each panel marking the information actually planted. **Left:** 1.151 nats planted in feature space, where it must survive tokenization — the curve never leaves zero, and the forecast-state probe stays below 0.05 at all 40 evaluation points. **Centre:** 0.9003 nats planted directly in token space, bypassing the tokenizer — the identical backbone at the identical budget recovers 0.8496 nats, 94.4% of what was planted, crossing the detection threshold at step 2,000. **Right:** the original feature-space plant, after the tokenizer interface is re-specified (Section 3.6) — recovery returns. Model capacity was never the constraint; the interface was.

and we report the intermediate failures because they are what establishes that.

(i) A pointwise fine encoder. The fine subtoken becomes a per-bar encoding of bar t ’s own features, produced by a branch with no cross-bar mixing, while the coarse subtoken retains the contextual causal encoder. Both properties are carried in one token pair; bits per bar, the vocabulary split and the cell design are untouched. Alone this changed nothing — legibility 0.5202, at chance. The optimizer routed the state through the contextual coarse smear and spent the fine bits elsewhere.

(ii) A per-bar bottleneck decode leg. A pointwise decode head must reconstruct bar t from bar t ’s fine code alone, with zero cross-bar access, and its loss is added to the training objective. This trains — the dimensions that win reach point-decoder correlation above 0.9 — but it does not select *which* dimensions win: legibility 0.5136 and 0.5037 at canonical width, and 0.5014, 0.4995, 0.5182 across three seeds on the entropy-calibrated fixture, a 3/3 failure against the 0.90 threshold. That failure is the measurement plotted in Figure 2: the leg works, and the objective still spends its budget on variance and covariance.

(iii) Class weighting, with objective separation. The bottleneck loss weights the six microstructure dimensions as a class by a factor λ , and the window reconstruction losses receive the fine channels detached, so the fine encoder is shaped exclusively by the per-bar objective. Detachment is load-bearing, and was measured to be: with the window loss attached the per-bar weighting is inert at every λ — per-dimension receipts are identical from $\lambda = 2$ to $\lambda = 10^6$ — because the window objective re-creates the smearing incentive as fast as the leg removes it.

λ is calibrated, not chosen. Across 26 trials no (λ, β) pair cleared 0.90 on all three seeds, which places the threshold inside the instrument’s seed-noise band at the achievable ceiling, while confirming that the state is carried (signal-dimension point-decoder correlation 0.77–0.96, microstructure class 0.86–0.92, return preserved above 0.96). The acceptance rule was therefore restated over the three calibration seeds as mean ≥ 0.90 and every seed ≥ 0.85 , and λ set to the smallest searched value clearing it: $\lambda = 2$ gives (0.8365, 0.8594, 0.8592), mean 0.8517 — fail; $\lambda = 3$ gives (0.9060, 0.9142, 0.9000), mean 0.9067, minimum 0.9000 — pass. Re-running through the pinned construction path reproduces mean 0.9047, minimum 0.8839, bit-identically to the direct construction at fixed thread count.

We state plainly that this is an acceptance threshold restated after seeing calibration results, and why it is not the failure that description usually names. The restatement was forced by a measured property of the instrument rather than chosen to admit a preferred value: 0.90 sits inside the calibration instrument’s own seed-noise band of roughly ± 0.03 at the achievable ceiling, so a fixed single-seed threshold at exactly that value tests the noise as much as the architecture, and the restated form — a mean clause plus a per-seed floor — is the standard response to that situation rather than a loosening of it. It occurred entirely on the synthetic calibration fixture, at a point when the real-data gate had never been executed and no real microstructure had been tokenized. And it did not propagate: the standing gate that governs expenditure (Section 3.7) requires ≥ 0.90 on *every one* of the six microstructure dimensions, with no mean clause and no floor, and nothing downstream of the calibration was selected on the restated form. $\lambda = 3.0$ is pinned in the conformance surface, applies identically to both quantizers and all input arms — so it cannot differentially favour any cell — and leaves bits per bar, the cell matrix and every evaluation instrument unchanged.

3.7 Verification, and the gate that carries it to real data

The intervention is verified by repeating Section 3.3 exactly: generator, seeds, stream size, plant strength, lag, budget, schedule and validation protocol unchanged, the tokenizer architecture the only difference. The forecast–state correlation crosses the detection threshold at step 2,000 and reaches 0.9407, with a maximum of 0.9414. The planted arm’s final validation NLL is 11.8478 against the noise arm’s 13.1859, a difference of -1.3381 nats where Section 3.3 gave $+0.0134$; the schedule completes without spikes and both arms stay in the sub-epoch regime. Per-bar legibility under the new tokenizer is 0.8990 on the planted arm and 0.9084 on the noise arm, where Section 3.5 measured 0.5135. We report Δ_{val} as the pre-registered decision statistic and not as an extraction fraction: each arm trains its own tokenizer, so the arm difference is not calibrated against the 1.151-nat plant in the way Section 3.4’s control is calibrated against its own H_0 . Figure 4 places every stage of the intervention against the gate.

A mechanism established on a fixture is a hypothesis about the real data, so we convert it into a gate rather than an expectation. After first-stage training and *before any second-stage expenditure*, each of the six microstructure dimensions must be linearly recoverable from bar t ’s own token pair at logistic sign-accuracy ≥ 0.90 on the run’s real training stream; failure raises an exception and halts the run. The probe is a logistic regression on the one-hot digit decomposition of the coarse and fine codes, trained for 300 Adam steps at fixed seed on an 80/20 split, deterministic given its inputs. Sign accuracy is well defined for the magnitude-shaped dimensions because every microstructure feature enters z -scored, so the sign is above-versus-below the causal rolling mean, and per-dimension base rates are recorded so that residual class imbalance is visible. The sample is stratified by symbol with the split blocked in time *within* each symbol, which preserves both properties that matter: full universe coverage, and no near-duplicate neighbours across the split. Two defects in this gate were found and closed before it was relied upon, both of the form “a check that cannot fail is not a check”; they are documented with their fixes in Appendix C.

The gate’s first execution on real microstructure is a named checkpoint with a pre-written adjudication: if the real dimensions cannot clear the threshold under the calibrated architecture, the run stops and reports with per-dimension receipts. It is not a threshold that may be moved. We record the expectation before the fact. Real trade-flow imbalance has the low-variance, weakly-covariant profile that Figure 2 shows being evicted, and the correction was calibrated on a synthetic plant. **A gate failure on real data would be the real-data confirmation of the mechanism** — simultaneously a blocker for the ablation and the strongest available evidence for the claim of Section 3.1. Both outcomes are reportable, and neither would be a surprise.

3.8 Scope of the claim

Four limits, restated in Section 7. First, everything above is measured on a *synthetic fixture*, not on real microstructure; the fixture’s non-signal structure is entropy-matched to the real data and its signal dimension is a controlled construction, which is what makes the recovered quantity exactly computable and is also what stops it being a measurement of markets. The mechanism is therefore not this paper’s headline claim, and is not elevated to one under any experimental outcome. Second,

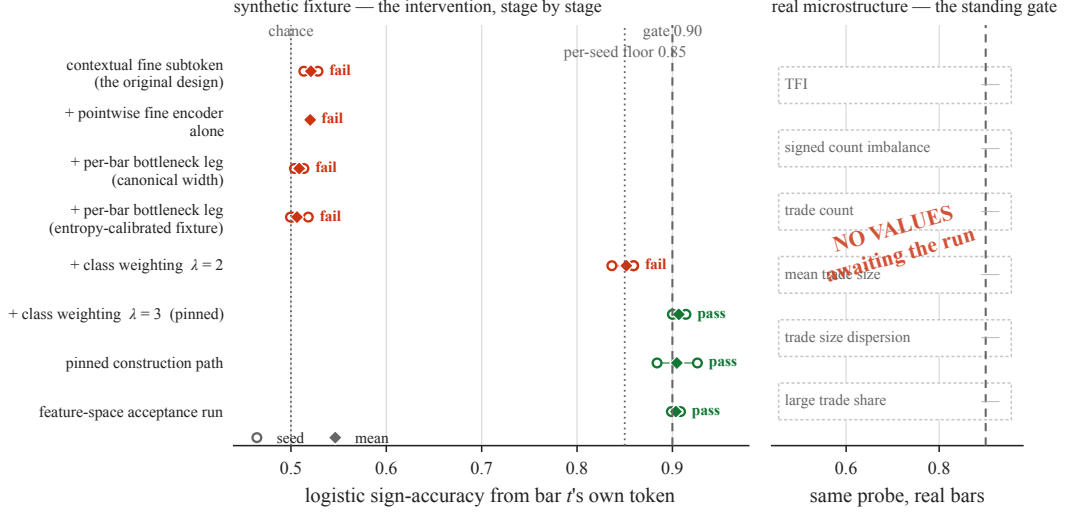


Figure 4: Only the third change moves per-bar legibility, and the threshold sits at the achievable ceiling. *Left:* logistic sign-accuracy of a microstructure dimension from bar t 's own token, across the intervention. Open circles are seeds and the diamond is their mean; the pre-registered acceptance rule over the three calibration seeds is mean ≥ 0.90 and every seed ≥ 0.85 , so both thresholds are drawn. The original contextual design, the pointwise encoder alone, and the pointwise encoder with the bottleneck leg all sit at chance — the leg trains, but does not decide which dimensions win. Class weighting at $\lambda = 2$ still fails; $\lambda = 3$ clears, and the pinned construction path reproduces it. Note how little headroom exists above the gate: this is the achievable ceiling, not a comfortable margin. *Right:* the same probe applied to real microstructure by the standing gate, which runs after first-stage training and before any second-stage expenditure. **[PH: no values drawn]** — the panel shows six empty slots rather than placeholder bars, because bars near the gate would imply a passing result nobody has measured. Six numbers drop in from the gate's own receipt, and the gate halts the run if any of them falls below the rule.

we demonstrate the effect for one reconstruction objective, one quantizer family and one class of encoder: we show that *this* objective allocates by variance and covariance, not that no reconstruction objective can do otherwise — and Section 3.6 is itself a counterexample to the strong form, being a reconstruction objective that carries the evicted channel because it was given an explicit term to do so. Third, the 94.4% of Section 3.4 is calibrated against a single matched reference run and inherits whatever variation that run carries. Fourth, λ is a hyperparameter fitted to a synthetic proxy for the real allocation problem; it is pinned pre-data and applied identically across every arm, so it cannot favour any cell, but the gate of Section 3.7 exists precisely because the proxy might not transfer.

A Eviction measurement, per dimension

Table 2 gives the per-dimension point-decoder correlations summarized in Table 1. Marginal standard deviations are implied by each dimension’s predict-the-mean baseline through $\mathbb{E}[X] = \sigma\sqrt{2/\pi}$, and confirm the fixture’s construction: every dimension except the return is unit-variance to two decimal places, so the return is the only high-variance channel and the signal dimension is distinguished from the fillers by dependence structure alone.

Table 2: Point-decoder correlation for every fixture dimension, three seeds. Recovery of a dimension’s value at bar t from bar t ’s own fine code. The eleven fillers are coupled by an AR(1) chain across dimensions at $\rho = 0.7993$; the signal dimension is independent of everything. Seed-to-seed variation within a class is at most 0.006, while the gap between the filler class and the signal dimension is more than 0.80.

dim	role	marginal sd	seed 0	seed 1	seed 2
0	return (high variance)	3.15	0.9818	0.9815	0.9823
1	filler	1.00	0.8931	0.8903	0.8920
2	filler	1.01	0.9128	0.9113	0.9109
3	filler	1.00	0.8523	0.8509	0.8565
4	filler	1.00	0.8261	0.8256	0.8319
5	filler	1.00	0.8793	0.8808	0.8816
6	filler	1.01	0.9120	0.9166	0.9101
7	filler	1.00	0.8793	0.8790	0.8738
8	filler	1.01	0.8258	0.8235	0.8242
9	signal (independent)	1.00	0.0140	0.0011	0.0053
10	filler	1.01	0.8513	0.8486	0.8514
11	filler	1.00	0.9179	0.9188	0.9171
12	filler	1.01	0.8969	0.8975	0.8940

Four dimensions clear a point-decoder correlation of 0.90 in each of the three seeds, which is the number the capacity arithmetic of Section 3.1 anticipates: a 10.26-bit fine budget cannot support thirteen dimensions at the fidelity that a 0.90 sign accuracy requires.

B Token-space plant: construction and calibration

The control of Section 3.4 requires a plant in token space whose information content is known exactly, is large enough to be detectable under the pre-registered branch condition (0.7–1.1 nats), and does not perturb the carrier’s marginal distribution enough to be detectable by that route instead.

A first design, retired. The natural construction — a deterministic translation of one digit by an amount determined by another — cannot carry enough information on this carrier. The carrier’s coarse digit marginals are already near-uniform: measured entropies of 2.3938, 2.1796 and 2.1873 nats against maxima of $\ln 11 = 2.3979$ and $\ln 9 = 2.1972$, i.e. 99.2–99.8% of maximum. Under near-uniform marginals a translation plant is capped at 0.2769 nats, below the required band, so it was retired as information-poor by construction rather than tuned.

The adopted design. The target digit is redrawn from a discretized Gaussian on the level grid, centred monotonically on the source level, with width σ_{plant} solved so that the exact mutual information lands in band. Table 3 gives the calibration receipt at solution and the same quantities recomputed over the full training stream. The closed-form and Monte-Carlo estimates agree to 9×10^{-4} at calibration and to 2×10^{-5} over the full stream; the marginal KL between the planted and carrier target distributions is 0.005 and the entropy shift is -0.005 nats, so the plant is not detectable from the marginal alone; and an unplanted control run returns a rank correlation of -8.8×10^{-5} , confirming the probe reads zero when there is nothing to read.

C Two defects in the legibility gate

Both defects are instances of the same failure — a check that cannot fail is not a check — and both were closed before the gate was relied upon. They are recorded because the gate is a hard stop on

Table 3: Token-space plant, calibration and full-stream receipts. The plant is solved to a target information content rather than tuned to a result; the oracle ceiling is the rank correlation an optimal predictor of the planted rule would achieve, and bounds what any model can reach.

quantity	at calibration	full stream
plant width σ_{plant}	1.0077	—
mutual information, closed form (nats)	0.9000	0.9003
mutual information, Monte-Carlo plug-in (nats)	0.9009	0.9003
closed form – Monte-Carlo	9.1×10^{-4}	2.0×10^{-5}
marginal KL, planted vs. carrier	0.00476	0.00480
entropy shift ΔH (nats)	−0.0054	−0.0052
oracle rank-correlation ceiling	0.926	—
unplanted control, rank correlation	-8.8×10^{-5}	—

expenditure, and because the second one was material.

A skipped dimension could not lower the verdict. A dimension with too few unmasked bars to measure took a code path that continued the loop *before* the pass flag was accumulated. A gate run in which all six microstructure dimensions were too thin therefore returned `pass: True` having measured nothing, and a run with five thin and one passing dimension returned `pass: True` on a gate whose stated contract is all six. Both were reproduced before the fix. “We could not measure the channel this gate exists to measure” is now a third outcome, and it halts.

The probe read one symbol of two hundred. The probe took a fixed 150,000-row head of a symbol-ordered concatenation with a contiguous train/validation cut. On the production data that window spans *one of 200 symbols*, so the check governing all second-stage expenditure was being made on a single instrument. The naive repair — shuffling before the split — would have broken the other property that matters, since one-minute bars are autocorrelated and an interleaved split puts near-duplicate neighbours on both sides. The fix is a per-symbol stratified split that is blocked in time *within* each symbol, which preserves both coverage and blocking.

The materiality of the two legs differs, and we measured rather than asserted it. On the production data the minimum per-dimension unmasked bar count across all 200 symbols is 277,758 against the gate’s 10,000-bar floor, with zero symbols below it, so the all-thin case cannot arise here: the first defect was real but immaterial on this data. The second was severe.

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